

Advanced Electronic Circuits Spring 26

Hayden Fuller

HW8

P1)

An op amp circuit is built to have a return ratio as given below where β is purely resistive but not necessarily known. $T(jf) = 10^5 / ((1 + jf/5)(1 + jf/10^5)(1 + jf/10^6))$

(a) What is the gain margin for this circuit? Do not estimate.

$$\angle T(jf) = -\arctan(f/5) - \arctan(f/10^5) - \arctan(f/10^6)$$

$$-180 = \angle T(jf) = -\arctan(f_{-180}/5) - \arctan(f_{-180}/10^5) - \arctan(f_{-180}/10^6)$$

$$f_{-180} = 316.236\text{kHz}$$

$$GM = -20\log_{10}|T(jf_{-180})| = -20\log_{10}|T(316\text{kHz})| = -20\log_{10}(0.45452) = 6.84893\text{dB}$$

(b) What is the phase margin for this circuit? Do not estimate.

$$|T(f_x)| = 1 = |10^5 / ((1 + jf_x/5)(1 + jf_x/10^5)(1 + jf_x/10^6))|$$

$$f_x = 210.203\text{kHz}$$

$$\angle T(f_x) = -166.428^\circ$$

$$\phi_m = 13.572^\circ$$

(c) How much should β be increased or decreased (multiplied by or divided by) to get a phase margin of 45° ?

$$\angle T(f_{x_2}) = -135 = -\arctan(f/5) - \arctan(f/10^5) - \arctan(f/10^6)$$

$$f_{x_2} = 84.438\text{kHz}$$

$$T(210.203\text{kHz}) = 1$$

$$T(84.438\text{kHz}) = 4.508$$

$$\beta \text{ divided by } 4.5087 \text{ (multiplied by } 0.2218)$$

(d) How much could β be increased or decreased (multiplied by or divided by) before the circuit becomes unstable?

$$\angle T(f_{x_3}) = -180 = -\arctan(f/5) - \arctan(f/10^5) - \arctan(f/10^6)$$

$$f_{x_3} = 316.236\text{kHz}$$

$$T(210.203\text{kHz}) = 1$$

$$T(316.236\text{kHz}) = 0.4545$$

$$\beta \text{ divided by } >0.4545 \text{ (multiplied by } <2.200)$$

P2)

An amplifier is being designed that has a DC gain of 200 kV/V, and two high-frequency poles at 100 kHz and 50 MHz. Note that in its current design, it is not a constant gain bandwidth product amplifier.

(a) What is the maximum value for β such that no gain peaking occurs? Assuming the system corresponding circuit is a non-inverting amplifier, what is the minimum closed loop gain?

$$A_0 = 200k$$

$$f_1, f_2 = 100k, 50M$$

$$A(jf) = A_0 / ((1 + jf/f_1)(1 + jf/f_2)) = 200k / ((1 + jf/100k)(1 + jf/50M))$$

$$Q > 1/\sqrt{2} = f_0 / (f_1 + f_2)$$

$$f_0 = 35.426 \text{ MHz}$$

$$f_0^2 = (f_1 * f_2) A_0 B$$

$$B = f_0^2 / (f_1 * f_2 * A_0) = 0.001255$$

$$B < 0.001255$$

$$A = 1/B > 796.810$$

(b) A capacitor is to be added within the amplifier design (for example, see Book Figure 5.1) to add a low frequency pole in the amplifier open loop response. At what maximum frequency does this pole need to be placed for $\beta = 1$ to result in $\phi_M = 60^\circ$?

$$A(jf) = 200k / ((1 + jf/100k)(1 + jf/50M)(1 + jf/f_3))$$

$$T(jf) = A(jf)$$

$$\angle T(jf) = -120 = -\arctan(f/100k) - \arctan(f/50M) - \arctan(f/f_3)$$

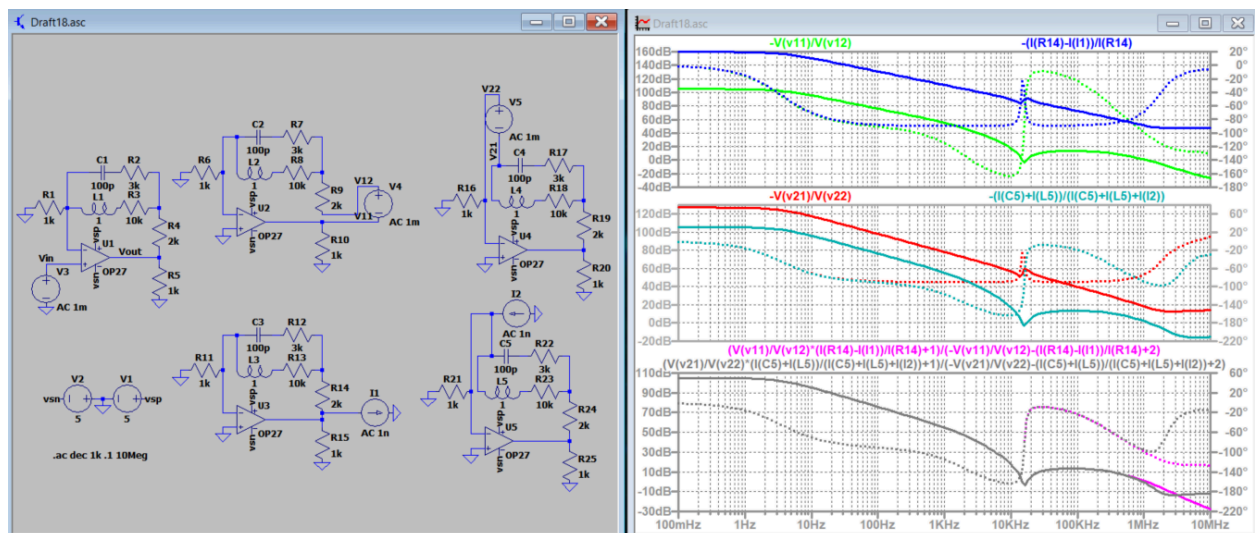
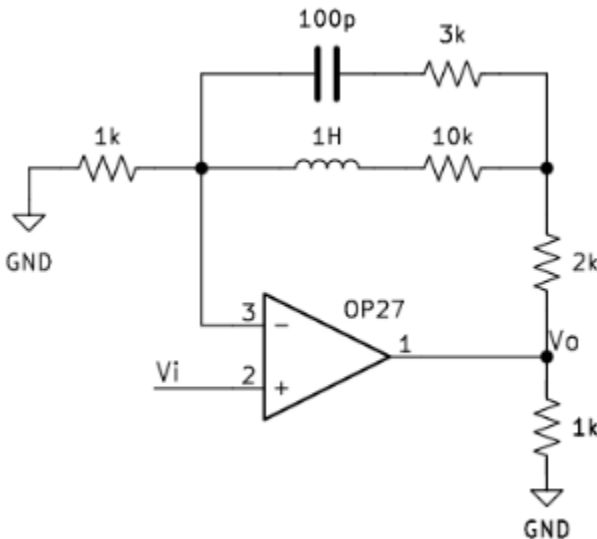
$$|T(jf)| = 1 = 200k / (|1 + jf/100k| |1 + jf/50M| |1 + jf/f_3|)$$

$$f = 57.7582 \text{ kHz}$$

$$f_3 = 0.33223 \text{ Hz}$$

P3)

Recreate this circuit in LTspice and measure its $T(jf)$ in two different locations via the Middlebrook technique. For each location, perform both voltage and current injection and determine if one or both are needed to determine $T(jf)$ and explain why. Perform the characterization of $T(jf)$ over the frequency range of 1 Hz to 10 MHz/ Verify that the resultant $T(jf)$ is functionally equivalent between the two test points (excluding possible simulation inaccuracies, similar to what was shown in lecture). The amplifier is powered by $\pm 5V$ supplies. Provide a screenshot of the built circuit showing the use of the Middlebrook technique at each point and any applicable output plots.



Only one injection is needed, we can see the node 1 voltage injection, node 2 current injection, and final solution are all approximately matching. Voltage injection works at nodes that are voltage driven with low impedance (such as the output of an op amp) and current injection works at nodes that are current driven with high impedance (such as the input of an op amp) Both plots are equivalent up to about 1MHz, and past that is assumed due to simulation inaccuracies.

P4)

Consider a two-pole non-GBP amplifier with the open loop gain given below, use Rate of Closure analysis and the axes below to answer each subquestion. $a(jf) = 10^5 / (1 + jf/10) (1 + jf/10^4)$

(a) Begin by plotting the straight-line estimated representation for $|a(jf)|$. Label the axes.

100k = 100dB

$f_1 = 10\text{Hz}$

$f_2 = 10\text{kHz}$

(b) Estimate the cross-over frequency, f_x , and phase margin for each of the three constant β values: $\beta = 0.1$, $\beta = 1/10^2$, $\beta = 1/10^4$. Label each corresponding line on the graph.

$1/0.1 = 10 = 20\text{dB}$

$f_x = 30\text{kHz}$

ROC = 40dB/dec

phase margin = 0°

$1/0.1 = 100 = 40\text{dB}$

$f_x = 10\text{kHz}$

ROC = 30dB/dec

phase margin = 45°

$1/0.0001 = 10\text{k} = 80\text{dB}$

$f_x = 100\text{Hz}$

ROC = 20dB/dec

phase margin = 90°

(c) Assume β also contains a pole, such that $\beta = 1 / (10^3 (1 + jf/f_p))$. What is the minimum value for f_p such that the circuit is stable?

$1/B_0 = 1\text{k} = 60\text{dB}$

$f_p > 1\text{kHz}$

